

# Sequences and series of functions

Math 163 • Prof. Chowdhury

The next page has 14 ideas as boxes, with 15 numbered arrows between them and nothing written on any arrow. An arrow is a relationship between two ideas, and not always an implication. Sometimes the first box gives you the second. Sometimes it doesn't, and there's a counterexample. Sometimes each gives the other. And sometimes it does give it, but on hypotheses you wouldn't have guessed.

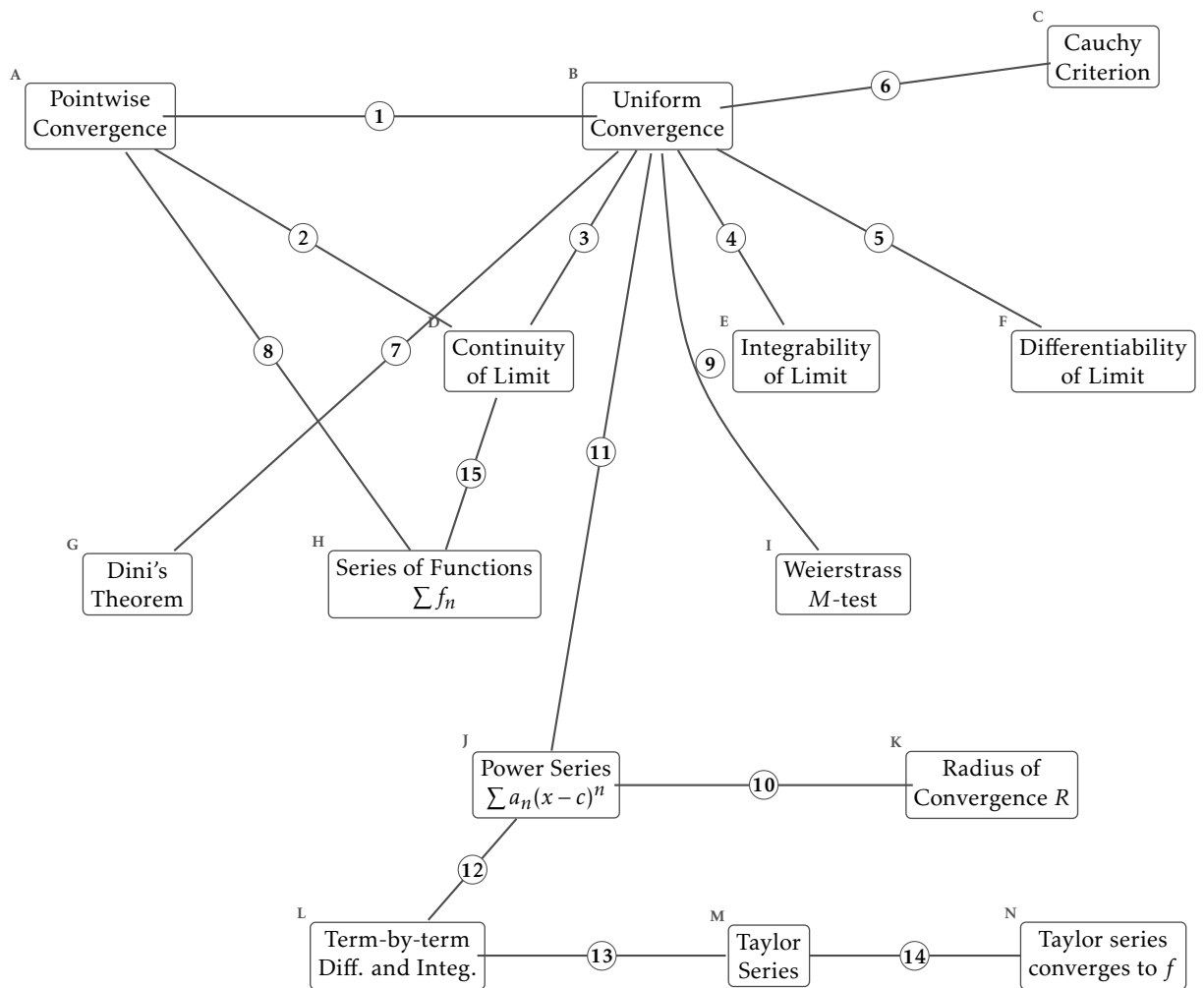
For each arrow, circle which of those four it is, and describe the relationship in one sentence. Name the theorem where there is one, and say what goes wrong where there isn't.

## What each box means

- A. Pointwise Convergence.** For each  $x$  and each  $\varepsilon > 0$  there is an  $N$  depending on both, with  $|f_n(x) - f(x)| < \varepsilon$  for  $n \geq N$ .
- B. Uniform Convergence.** For each  $\varepsilon > 0$  there is an  $N$  depending only on  $\varepsilon$ , with  $|f_n(x) - f(x)| < \varepsilon$  for  $n \geq N$  and every  $x$  at once.
- C. Cauchy Criterion.**  $\{f_n\}$  converges uniformly exactly when for each  $\varepsilon > 0$  there is an  $N$  with  $|f_m(x) - f_n(x)| < \varepsilon$  for all  $m, n \geq N$  and all  $x$ .
- D. Continuity of Limit.** Each  $f_n$  continuous and  $f_n \rightarrow f$  uniformly gives  $f$  continuous.
- E. Integrability of Limit.** Each  $f_n$  integrable and  $f_n \rightarrow f$  uniformly gives  $f$  integrable, with  $\lim \int f_n = \int \lim f_n$ .
- F. Differentiability of Limit.**  $f_n \rightarrow f$  pointwise together with  $f'_n \rightarrow g$  uniformly gives  $f' = g$ . It is the derivatives that have to converge uniformly.
- G. Dini's Theorem.**  $f_n \rightarrow f$  pointwise on  $[a, b]$ , monotonely, with every  $f_n$  and  $f$  continuous, gives  $f_n \rightarrow f$  uniformly.
- H. Series of Functions**  $\sum f_n$ .  $\sum f_n$  converges when its partial sums  $S_N(x) = \sum_{k \leq N} f_k(x)$  do. A series of functions is a sequence of functions.
- I. Weierstrass M-test.** If  $|f_n(x)| \leq M_n$  for every  $x$  and  $\sum M_n < \infty$ , then  $\sum f_n$  converges uniformly and absolutely.
- J. Power Series**  $\sum a_n(x - c)^n$ . A series of functions whose  $n$ th term is a monomial of degree  $n$  centred at  $c$ .
- K. Radius of Convergence**  $R$ .  $\sum a_n(x - c)^n$  converges absolutely for  $|x - c| < R$  and diverges for  $|x - c| > R$ . The endpoints are checked one at a time.
- L. Term-by-term Diff. and Integ..** Inside  $(c - R, c + R)$  a power series can be differentiated and integrated term by term, and the result has the same radius  $R$ .
- M. Taylor Series.**  $\sum f^{(n)}(a)(x - a)^n/n!$ . The coefficients are forced by differentiating the power series at its centre.
- N. Taylor series converges to  $f$ .** The Taylor series converges to  $f(x)$  exactly when the remainder  $R_N(x) = f(x) - P_N(x)$  tends to zero.

## Counterexamples and benchmarks

- $x^n$  on  $[0, 1]$  takes continuous functions to a discontinuous limit, pointwise.
- $\sqrt{x^2 + 1/n^2} \rightarrow |x|$  uniformly, and the limit is not differentiable at 0.
- $(1/n) \sin(n^2 x) \rightarrow 0$  uniformly while its derivatives diverge.
- $x/(1 + n^2 x^2) \rightarrow 0$  uniformly, and the derivative of the limit is not the limit of the derivatives at 0.
- $e^{-1/x^2}$  extended by 0 is infinitely differentiable at 0 with every Taylor coefficient zero, so its Taylor series converges and not to it.



Every arrow is drawn the same way on purpose. Deciding the kind is half of each answer.

## The arrows

1. gives / fails / both ways / gives, other hypotheses

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2. gives / fails / both ways / gives, other hypotheses

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3. gives / fails / both ways / gives, other hypotheses

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4. gives / fails / both ways / gives, other hypotheses

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5. gives / fails / both ways / gives, other hypotheses

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6. gives / fails / both ways / gives, other hypotheses

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10. gives / fails / both ways / gives, other hypotheses

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11. gives / fails / both ways / gives, other hypotheses

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12. gives / fails / both ways / gives, other hypotheses

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13. gives / fails / both ways / gives, other hypotheses

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14. gives / fails / both ways / gives, other hypotheses

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15. gives / fails / both ways / gives, other hypotheses

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## One last question

Which single arrow carries the most of this chapter? Say why in one sentence. If your answer is the one arrow whose hypotheses are not what you expected, say what you expected instead.

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